Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies Donovan	AO1.2	B1	First Outlier Donovan
	Infers a reason for Donovan • a data entry error has been made • Donovan gets nervous/stressed under exam conditions and performed poorly in most recent test • Donovan is not a very good player despite a lot of practice • Just started playing so practised longer hours but performed poorly in exam • External factors/illness Accept other reasonable reason linking Donovan's practice time to their performance in the exam.	AO2.2b	E1	Reason A data entry error has been made (should be 5 not 50)
	Identifies Collins	AO1.2	B1	
	Infers a reason for Collins • a data entry error has been made • naturally very good piano player so does little practice • 'Lucky' test score Accept other reasonable reason linking Collins's practice time to their performance in the exam.	AO2.2b	E1	Second outlier Collins Reason Naturally very able student.
(b) (i)	Describes correlation correctly, at least strong positive. Accept non-linear correlation, but do not accept numerical value to indicate strength	AO2.5	B1	Strong Positive Correlation
(ii)	Interprets correlation in	AO3.2a	E1	Students who complete

context (as given in typical solution OE) Do not accept the better you do in the exam the more you practised			more practice perform better in the exam
			Total 6 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Indicates that 'Semi-skimmed milk' is part of the whole' Skimmed milks' category.	AO2.4	E1	In Figure 2, the vertical axis represents 'Skimmed milks' whereas in Figure 1 the vertical axis represents 'Semi-skimmed milk' that is just one part of the 'Skimmed milks' category. Hence, for each region, the recorded values for 'Semi-skimmed milk' will always be lower that the ones for 'Skimmed milks'.
(b) (i)	States: **strong positive** correlation **between** purchases of **'Semi-skimmed milk' and 'Liquid wholemilk, full price'** for most regions.	AO2.5	B1	Regions A, B, D, F, G, H and J indicate strong positive correlation between purchases of 'Semi-skimmed milk' and 'Liquid wholemilk, full price'.
	Identifies regions C and E as outliers	AO3.2b	E1	Regions C and E do not follow the same pattern as the other regions in terms of purchases of skimmed milk and wholemilk or regions C and E can be excluded as they appear to be outliers.
ALT	Allow 1 mark if candidate states: **weak negative** correlation between **consumption of 'Semi-skimmed milk' and 'Liquid wholemilk, full price'**.	AO2.5 AO3.2b	B1 E0	

(ii)	States that there is evidence of negative correlation between purchases of 'Mineral or spring water' and 'Skimmed milk'.	AO3.1b	B1	Scatter graph indicates evidence of negative correlation between consumption of 'Mineral or spring water' and 'Skimmed milks'.
	The correlation is between water and milk purchases within the regions and not between purchases made by individual people so Bilal's claim is not proven	AO2.3	E1	Bilal's claim is not supported because the correlation in figure 2 is between water and milk purchases within the regions and he is assuming that individuals will follow a similar pattern but the data does not tell us anything about individuals, it refers to average purchases made within the nine regions.
(c)	Identifies London	AO2.2b	B1	London
	Gives valid reason based on knowledge of the LDS	AO2.4	E1	Having studied the LDS I have identified a clear trend: London is a frequent outlier in most categories
				Total 7 marks

Q3.

(a) $r = 0.6$ to 0.98

AWFW (≈ 0.8)

If answers are not labelled, assume order is (a) then (b)

B1

(b) $r = -0.5$ to 0.02

AWFW (≈ –0.3)

Accept answers as ranges if and only if contained entirely within given ranges

Eg: (a) 0.7 to 0.9 ⇒ B1

(b) –0.6 to –0.4 ⇒ B0

B1

[2]

Q4.

(a) $-0.95 \le$ **Value** ≤ -0.50

Actual value is –0.80

B2

(–1 < **Value** < 0)

Accept range only if within that given

(B1)

(b) –0.10 ≤ **Value** ≤ + 0.10

Actual value is +0.005

B2

(–0.20 ≤ **Value** ≤ 0.20)

Accept range only if within that given

(B1) 4

[4]

Q5.

(a) Probably incorrect

CAO

B1

Expect height to **increase** with age
Expect **positive** value

Or equivalent

B1 2

(b) Definitely incorrect

CAO

B1

Value of r cannot exceed 1

Or equivalent

B1 2

(c) Probably correct

CAO

B1

Expect weight to increase with age

Or equivalent

B1 2

[6]

Q6.

(a) The takings appear to increase slightly as the air temperature increases

OE

B1

Weak positive (linear) correlation between air temperature and takings

Comments on ranges of values of x and y $\Rightarrow$

B0

One (or two) unusual results

OE

B1

2

(b) Monday 10

CAO; accept point (4, 312)

B1

1

(c) Temperature at another time

Or a sensible alternative

Number of other/competing stalls

Month/time of year

Number of customers $\Rightarrow$ *E0*

Rainfall/snow

Weather $\Rightarrow$ *E0*

Publicity

Population of town $\Rightarrow$ *E01*

E1

1

[4]

Q7.

(a) 0.0

0.0 (–0.2 ~ 0.2)

B1

(b) –0.8

negative and > –1

B1

magnitude 0.6 ~ 0.98

B1

(c) –0.8

negative and > –1

M1

magnitude 0.4 ~ 0.98

A1

5

[5]

Q8.

1. probably incorrect (B)- would expect negative correlation coefficient

Probably incorrect

B1

Negative expected

E1

2. Definitely incorrect (C)-*r* cannot exceed 1

Definitely incorrect

B1

Cannot exceed 1

E1

3. Plausible (A) - probably both related to population of town

Plausible

B1

Related to population of town

E1

[6]

Q9.

(a) *y* increases rapidly with *x* for small values of *x* and then levels out.

y increase with x

E1

Relationship changes

E1

Description of change

E1

3

(b) Not appropriate because relationship nonlinear

Non-linear

M1

Not appropriate

A1

2

[5]

Q10.

(a) –0.9

$-1 \leq r < 0$

$0.7 \leq$ magnitude < 1

B1 B1

2

(b) 0.7

$0 < r \leq 1$

$0.2 \leq r \leq 0.8$

(If second B1 awarded in (a) this magnitude must be smaller)

B1 B1

2

(c) –0.8

$-1 \leq r < 0$

$0.2 \leq$ magnitude ≤ 0.9

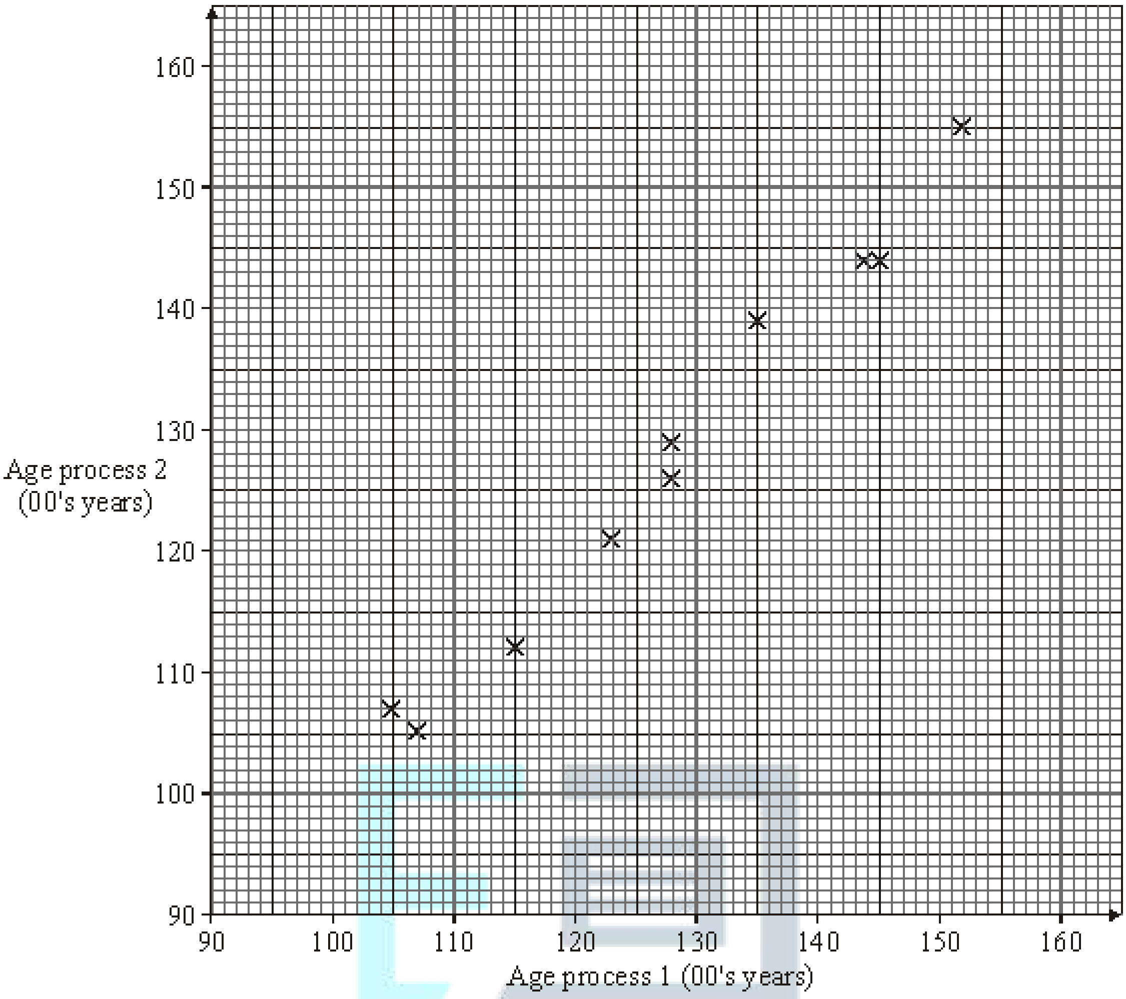
M1 A1

2

[6]

Q11.

(a) (i) ***See graph below***



B1 axes, M1 attempt, A1 all appear correct by eye (allow one small slip)

B1 M1 A1

3

(ii) Comments – scatter diagram indicates a strong direct (linear) association between the ages given by the two processes.

B1 B1

2

(b) H_o $\rho = 0$

H_1 $\rho > 0$ 1 tail 1%

B1

test stat , r = 0.992 cv = 0.7155

For cv

B1

r > 0.7155 hence significant evidence exists to reject H_o.

There is sufficient evidence to suggest a direct association between the ages given by the two processes

For comparison ts /cv

M1 A1

4

[9]